

A Partisan Pandemic: How COVID-19 Was Primed for Polarization

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Abstract

Despite the universal impact of COVID-19, Americans in different political parties rapidly formed diverging attitudes about the pandemic. Although other matters of scientific concern had previously elicited partisan reactions, Americans had diverged on those issues over decades rather than months. In this paper, we review evidence on factors that led to polarization of previous scientific issues to explain why reactions diverged so quickly this time around. Secondary analyses of publicly available survey data reveal that partisan reactions to the pandemic were closely associated with trust in public health institutions and that the association between partisanship and trust increased over time. In contrast, an alternative hypothesis that conservative media use might explain polarization was not supported. The conflation between trust and partisanship appears to largely explain polarized reactions to COVID-19 over time.

Keywords: *polarization, trust in science, public health, coronavirus, partisanship*

Introduction

Shortly after the U.S. reported its first known COVID-19 related death on February 29, 2020, the country seemed poised to come together to address the impending threat of a global pandemic (CDC, 2020). On March 13th, President Trump declared a national public health emergency related to the spreading virus, a move supported by politicians of both parties. Even then, there were partisan differences in concern about the virus (Deane et al., 2021), though these may have reflected the fact that cases were detected in urban and more Democratic areas before they spread to more Republican and rural areas (cf. Allcott et al., 2020). By mid-April, however, Americans diverged not just in concern, but also in policy preferences—Republican groups challenged Democratic governors on what they considered overly broad emergency orders, organizing large protests in Lansing (MI), Richmond (VA), and St. Paul (MN), which were backed by then-president Trump (Haberman, 2020). By June, 76 percent of Democrats said they usually wore masks in stores versus only 53 percent of Republicans (Kramer, 2020). And what initially appeared to be a non-partisan public health issue had morphed into a highly-polarized one, affecting partisans’ attitudes and behaviors, as well as their support for various preventative measures and policies. Here, we examine two questions: why did responses toward COVID-19 polarize so quickly, and why did they polarize at all?

To understand what happened, we draw from research on divergence in Americans’ perceptions of other scientific issues that entered the policy domain. In general, explanations for scientific polarization rely on three basic mechanisms: (1) General differences towards science driven by political elites (Brulle, Carmichael, and Jenkins, 2012), (2) differences in trust towards public health institutions and officials (e.g., Mooney, 2005), and (3) differences in the information that different groups receive about scientific issues (e.g., Feldman et al., 2011). Solid evidence

bolsters each of these claims with respect to issues like anthropogenic climate change and belief in evolution (Miller et al., 2006). And each type of explanation has also been posited to explain diverging partisan reactions to the coronavirus pandemic (Deane et al., 2021). In this paper, we test how well key expectations from each hold up in explaining partisan reactions to COVID-19.

General Partisan Differences Driven by Political Elites

Although Americans across political parties agree more than they disagree when it comes to significant political issues, party elites often have strong, diverging views on those same issues (Fiorina, 2017). Despite these shared sentiments, a growing body of work evinces considerable partisan animosity (Iyengar, Lelkes, Levendusky, 2019), which frequently results in individuals reflexively adopting positions supported by in-party elites and opposing those offered by out-party elites (Banda & Cluverius, 2018). As elites stake out positions on various issues, this results in popular divisions on these same matters (Zaller, 1992). Elite-driven attitude formation is often associated with a lack of pre-existing policy opinions in the general populace. As opposed to deep, critical thought about an issue, elite positions can be used as a heuristic for individuals determining what their attitudes should be. And, since elites' stances tend to be both extreme and diametrically opposed, this process encourages polarization in people who adopt these attitudes about a given issue. While this is generally constrained to political issues, when an issue that should arguably be non-partisan becomes entangled in politics, polarization is likely to follow.

The rapid developments of the coronavirus pandemic provided a display of issue alignment and party bifurcation in real time. The World Health Organization was first made aware of the virus in December 2019, and the first confirmed case of COVID-19 in the United States was reported in January 2020 (CDC, 2021). It was therefore impossible for Americans to have formed attitudes about it before December, and unlikely that many did so before the disease began to

spread nationally in March. The speed with which coronavirus entered America's public consciousness meant that Americans had to quickly form opinions in a low-information environment, and may have looked to elites for cues.

Wearing masks in public quickly became a partisan symbol as Democratic elites promoted mask-wearing and Republican elites—most notably President Trump—dismissed it. Later, vaccination intentions followed the same route, with the president and other Republican elites providing at best tepid endorsements (Wise, 2021). This was true even though the CDC, which recommended masks, was led by Trump appointee Robert Redfield and the administration could claim credit for the rapid development of a vaccine through Operation Warp Speed. In a study exploring whether elite positions might impact public attitudes, Pink and colleagues (2021) found that when Republicans viewed vaccine endorsements from in-party elites, they were significantly more likely to indicate an intent to be vaccinated; at the same time, Republicans who viewed endorsements from Democratic elites became less inclined. Differing elite positions may partly explain differences in public attitudes and behaviors.

Trust Towards Public Health Institutions

A second account is that political parties may differ more generally in acceptance of science, as Mooney contended in his 2005 book, *The Republican War on Science* (Mooney, 2005). Although critics claimed that the book examined scientific issues that were particularly partisan in orientation (Berube, 2006), differences in deference to scientific information may nonetheless be real. Indeed, Gauchat's (2012) study examining broader trends in public trust in science and scientists between 1974 and 2010 confirmed a link between trust and ideology. And Krause and colleagues (2019) reported that trust differences were increasingly associated not just with ideology, but also with partisanship. Ironically, there is even evidence that Mooney's argument

may have been self-fulfilling, as mentions of a Republican “war on science” can lead conservatives to assert greater distrust in science (Hardy et al., 2019). Because trust in science is an important component of both belief in scientific information and responsiveness to that information, systematic scientific distrust could pose serious problems.

When science policy is a subject of political contention, even a scientific consensus can become politicized as individuals feel justified in ignoring it (Bolsen and Druckman, 2018). Those who don’t trust scientists may feel empowered to assert that scientific findings are likely wrong (Chinn and Pasek, 2020) or may be predisposed to rejecting the scientific consensus (cf. Pasek 2018; Lewandowsky, Gignac, and Oberauer, 2013a). Indeed, scientific rejectionism occurs most prominently when individuals are motivated to do so (because claims conflict with respondents’ identities or beliefs) and when they have little regard for science and scientists (Lewandowsky, Oberauer, and Gignac 2013b; Rutjens et al., 2017). These circumstances occur when religious conservatives encounter information about evolution and when Republicans are told about climate change.

Amid the novel coronavirus pandemic, the role of trusting public health institutions and scientists could be profound. Early in the pandemic, scientific information was rapidly changing (Cross, 2021) and had to compete with large flows of misinformation, disinformation, and conspiracy theories (Peters and Crynbaum, 2021). Concurrently, trust in public health authorities was intertwined with the attitudes Americans had about the pandemic and their behavioral intentions (**CITE APPC WORKING PAPER**). If Americans in different partisan groups had differing trust in these institutions, they might similarly diverge in their tendency to get accurate information about the pandemic.

Differences in Information Associated with Partisanship

When issues become a source of political contention, individuals in different groups increasingly encounter and process information in ways that can reinforce group differences. A growing body of work in communication has shown that individuals can and do seek information sources that reinforce their political viewpoints (Stroud, 2011), even if they typically do not do so exclusively (Garrett & Stroud, 2014). Recently, scholars have documented the emergence of a distinct conservative media ecosystem capable of generating and sustaining its own knowledge base rooted in conservative ideology (Jamieson & Cappella, 2008). One distinctive feature of conservative media is that it presents certain scientific issues in a dismissive manner, a phenomenon well documented for anthropogenic climate change (Feldman et al., 2012). Conservative outlets disproportionately presented skeptical or falsely balanced narratives about climate change; correspondingly, heavy conservative media viewers were less certain of global warming than those who consumed less (Hmielowski et al., 2014) and had more misperceptions about the state of scientific consensus surrounding climate change (Hamilton, 2016).

Similar patterns of questionable coverage were present on conservative media with respect to COVID-19. Conservative media viewers were more likely to distrust pandemic information and to think that COVID-19 health risks were being exaggerated (Motta et al., 2020). They were also prone to coronavirus-related conspiracy theories (Romer & Jamieson, 2020). Conservative pundits appearing on these outlets often claimed the virus was a fraud or a Chinese plot to harm the U.S. economy and sought to downplay the effects of the pandemic (Peters & Crynbaum, 2020). Messages received differed depending on the sources people viewed, but polarized presentations were common across conservative outlets (Chinn et al., 2020). Hence, exposure to conservative sources might be causally related to concern about coronavirus, mask wearing, and vaccination intentions.

The Current Study

In the current study, we examine differences in concern about coronavirus, mask wearing, and vaccination intentions by party over time. We then examine how well these gaps can be explained as a function of partisanship, trust in public health institutions, and conservative media use. By understanding the factors most closely associated with the observed differences, we can formulate more relevant strategies for addressing and combating future scientific polarization.

Method

Data

Data for the current study come from a secondary analysis of the Axios/Ipsos Coronavirus Index, a weekly survey of attitudes, beliefs, and behaviors about coronavirus. Most weeks, beginning on March 13th, 2020 and continuing through the period when this study was written, panelists at Ipsos KnowledgePanel® were invited by email to complete surveys about topics related to coronavirus. These panelists had been recruited via address-based sampling and were provided with a tablet and Internet connection if these were not already available. Individuals aged 18 and older living in sampled households were eligible to join the panel. After joining the panel, around 1000 individuals responded to each week's survey (range: 979-2054). Cumulative response rates for the study hovered around 3-4 percent (CUMRR3; see Callegaro and Disogra, 2008).

Analyses presented in this paper rely on 56,787 responses from waves 4 to 53 of the study (from April 4, 2020 to September 11, 2021). Results of each analysis are only presented across waves where consistent measures were available. Because respondent identifiers were not appended to the individual datasets, it is unclear exactly how many individuals were interviewed

over the field period (KnowledgePanel® contained around 60,000 during this timeframe, so some respondents were likely interviewed multiple times).

Outcomes measures were three questions about how concerned respondents were about coronavirus, how often respondents were wearing a mask when they left their homes, and how likely they were to get the first generation COVID-19 vaccine as soon as it was available. All outcomes were coded to range from 0 (for those least concerned, who never wear a mask, or who would not get the vaccine) to 1 (for those most concerned, who always wear a mask, and who would get the vaccine or had already gotten it). Measures for demographics (age, race, gender, education, urbanicity), partisan self-identification as a Democrat, Republican, or independent, trust in public health institutions (an index of two questions about trust in the CDC and public health officials), and conservative media use (selecting FOX News or Conservative Online News as one's primary source of news) were used in these models. Full measures can be found in Online Appendix A.

Analysis

To assess how partisanship, trust in public health institutions, and conservative media use each related to concern, mask wearing, and vaccination intentions, we generated two sets of models. One group was designed to understand the net correspondence between each predictor and outcome, after demographics were controlled. A second set of models was designed to understand the unique correspondence between each predictor and the outcomes after demographics and the other predictors were controlled. Both values were calculated by running regressions within each survey week and determining the difference in variance explained (ΔR^2)

between models with the target predictor and those that did not include the target predictor. Regressions for the individual weeks are presented in Online Appendix **B**.

To illustrate how relations between key predictors and outcomes changed over time, generalized additive models (GAM) predicted the value of each week's explained variance as a smoothed function of time. GAMs allow us to fit spline curves to these trends without overreliance on the differences in survey data from week to week.

Results

Partisan identity was a strong predictor of all outcomes. Compared to Republicans, Democrats and Independents were far more concerned about coronavirus, likely to wear masks, and likely to get vaccinated. In general, these differences grew over time, at least through the 2020 U.S. Presidential Election. Among Democrats, 32.0 percent said that they were very concerned about coronavirus in April of 2020 compared to 14.6 percent of Republicans, a difference of 13.4 percentage points; this increased to a maximum of 40.1 percentage points by September. Figure 1A shows trends in partisan concern over time, estimated using GAMs. Similar divergences can be observed in partisan trends for mask wearing (Figure 1B) and vaccination intentions (Figure 1C).

[Figure 1 - Trends in coronavirus related attitudes by partisan self-identification]

Though these partisan differences are glaringly present, a more nuanced understanding of *why* people adopt the attitudes they do is needed. For instance, some differences between Democrats and Republicans could be due to the different kinds of individuals who join the parties, so it helps to account for those differences when estimating the effect of partisan identities. Similarly, to understand how much of the partisan difference might be attributable to factors like

institutional trust and conservative media use, we investigate what role partisanship plays when these factors are controlled.

Party Differences in Variance Explained

To estimate the influence of partisanship beyond demographic differences and assess whether that influence was independent of other key predictors, we calculated the extent to which partisanship measures improved the fit of regression models predicting each outcome within each wave. This yielded a single parameter that could be plotted over time to see how the influence of various predictors changed. Figure 2 shows that partisan identification became an increasingly important discriminator of attitudes over the course of the campaign for all outcomes.

[Figure 2 - Party-Attributable R^2 for outcome variables]

Predicting concern about coronavirus, we find that a model with just demographics produces R^2 values ranging from .05 to .11 depending on the week examined. Models also including partisanship yielded values ranging from .18 to .27. By subtracting the demographics only estimates of variance explained from the ones including partisanship, we estimate the variance attributable to partisanship (ΔR^2) as ranging between .05 and .18 depending on the week, with the smallest attributable variation in early April of 2020, the largest gaps in November/December of 2020, and a shrinking difference thereafter (see blue lines in Figure 2A).

While the total variance explained increased when measures of trust in public health institutions and conservative media use were included (ranging from .26 to .37), the portion attributable to partisanship dropped (to between .01 and .11), suggesting that variance in these other measures overlapped significantly with partisan differences. The partisanship attributable variance is plotted with the dashed red line in Figure 2A. Here it becomes clear that the uniquely partisan component of concern was moderate before Election Day but diminished thereafter.

The difference in variance explained between the solid blue and dashed red lines in Figure 2A reveals the extent to which what initially appeared to be partisan differences could be attributed either to trust in public health institutions or conservative media use. Because the difference between those lines was always sizable and appeared to increase after the election, we can conclude the additional variables captured much of what initially presented as partisan differences, and that the extent to which partisan gaps could be explained by these factors was larger after the election than it had been before. That is, while the partisan difference remained, the initial gap was not solely about partisanship.

Patterns linking partisanship with variance explained were similar for mask wearing. The partisan gap in mask wearing also grew between April and the election and shrunk thereafter. Similarly, variance in mask wearing uniquely attributable to partisanship was much smaller after Election Day than beforehand (Figure 2B).

For vaccines, unlike for the other two outcomes, the overall partisan gap increased over time (blue line in Figure 2C). But again, most of this difference was not uniquely explained by partisanship. After accounting for institutional trust and conservative media use, the variance uniquely explained by party differences was minimal.

Differences in Variance Explained by Trust and Media Use

The collective impact of trust in public health institutions was relatively large for all three outcomes (Figure 3A-C). Unsurprisingly, individuals who expressed more trust in these institutions were far more likely to report concern about COVID, wearing masks regularly, and intentions to vaccinate. These patterns persisted even after controlling for partisanship and conservative media use. And the role of trust in public health institutions became increasingly important over time across all three outcomes. While the variance shared between trust in public

health institutions and partisanship for each of the outcomes was sizable, this accounted for only a moderate proportion of the overall variance explained by trust, even though it was responsible for most of the variance attributable to partisan identification, at least in the post-election period.

[Figure 3a-f Trust- and Media Use-Attributable R^2 for outcome variables]

Reliance on conservative media, in contrast, did relatively little to explain variance beyond demographics for any of the three outcomes, and accounted for almost no unique variance when partisanship and trust in institutions were included in the model (Figure 3D-F). These results contraindicate claims that patterns of media use explain why individuals in different parties diverged in the beliefs, attitudes, and behaviors regarding coronavirus.

When comparing the unique variance attributable to each set of predictors, a consistent pattern emerges (Figure 4). Before Election Day, differences in concern and mask wearing across partisan groups were predicted by partisanship and trust in public health authorities. Afterward, these differences were increasingly attributable to trust rather than to party. And differences in vaccination intentions appear to have been less a factor of party than of institutional trust.

[Figure 4a-c Aggregate Models for Outcome Variables]

Is the Partisan Difference Related to Elite Cues?

In additional analyses, we tested the possibility that the unique partisan differences that remained might be attributed to elite cues. To assess this, we examined whether partisan identification explained significant differences in outcomes after also controlling for trust in President Trump (see Appendix A). These results are presented with the dotted green lines in Figure 2. In line with the elite cues explanation, the unique influence of partisan identification was nearly eliminated for all three outcomes.

Impact of Partisanship on Trust?

Although global trust in public health institutions was stable over time, this stability was not present on an individual level. From around when President Trump contracted COVID-19, Republicans and Democrats began to diverge in their trust toward these institutions at roughly an equal magnitude (Figure 5).

[Figure 5 - Partisan Differences in Trust]

To test whether this relationship accounted for the finding that trust explains more variance than partisanship, we decomposed the variance explained by partisanship into four categories: variance explained only by partisanship, variance explained jointly by partisanship and trust (but not media), variance explained jointly by partisanship and media (but not trust), and variance explained jointly by all three measures (calculations in Online Appendix C). Figure 6 shows trends in each over time. For both concern (6A) and mask use (6B), the unique variance attributable to partisanship (solid blue line) is the strongest predictor at first. Over time, however, the shared variance between partisanship and trust became stronger shortly after the election. For vaccination intentions, this shared variance was consistently more predictive than unique partisan variance, though the degree varied over time (Figure 6C). Between when the Pfizer vaccine was approved and when it became available to all adults in April 2021, this shared variance became increasingly predictive, before starting to decline until around July 2021 when it becomes about equal to unique partisan variance.

Collectively, while the direct effects of partisanship decreased, trust became increasingly conflated with partisanship after the election. The shared variance between trust and partisanship suggests that some variance observed in the former may be driven by the latter, though the independent variance of trust still was more consequential. These results are in line with what we might expect if the influence of partisanship is partially mediated by trust.

[Figures 6a-c Shared Variance Plots]

Discussion

This study examined how partisanship, trust in public health institutions, and conservative media use related to Americans' concern about COVID-19, mask use outside of the home, and intentions to get vaccinated over time. For all outcomes except concern, trust explained the largest proportion of the variance. In the case of concern, trust and partisanship were equally important predictors before the 2020 election, but trust became more important afterwards. And in all models, conservative media use exhibited a negligible impact. Notably, the explanatory power of trust in public officials was not consistent over time, but rather increased, particularly after the 2020 election, in conjunction with a corresponding decrease in the influence of partisanship.

But while the story may in part reflect the declining salience of partisanship after the election, party divides did not simply disappear as trust became more important. Instead, starting around the time President Trump revealed his own COVID-19 diagnosis, the correspondence between partisanship and trust itself began to increase, with Republicans expressing less trust and Democrats reporting greater trust. This left the overall level of trust largely unchanged, even though the composition of those who were trusting had shifted notably. Hence, while trust appears to have been the most important reason that individuals were worried about coronavirus, masked up, or intended to vaccinate, the conflation of partisanship and trust over time implies a different mechanism.

The results, then, are compatible with stories about elite polarization as well as with broad differences in trust in science across political groups. The fact that trust and partisanship became more conflated over time may indicate a vicious cycle effect as well, where elite pushback against scientifically motivated policy has not only short-term, but also longer-term consequences. While

it is likely too late to depolarize COVID-19, interrupting this cycle may be important in addressing future scientific issues.

Evidence that this process was not driven by conservative media is important and interesting, though not particularly conclusive, given the measure employed in these data. Rather than asking for the frequency of media use, respondents were simply asked which source they consult *most* frequently. Absent data showing the degree of media use, our minimal results are far from definitive.

In addition, we caution that the results presented here are limited in critical ways. Although current literature would suggest that partisanship likely precedes trust in public health institutions, trust in these institutions maybe impacts partisanship, for instance. It is also possible that some other source of attitudes and information, perhaps related to the president, could be accounting for the shared variance observed. Unfortunately, many of the variables we consider already had strong relations before the study began, rendering our effort in disentangling them at best partial. Panel research assessing how individual attitudes change over time would clarify the ordering of causal relations and enable a more formal mediation analysis.

Despite these limitations, the results suggest that polarization around COVID-19 echoes some tropes from earlier cases of scientific polarization and highlights at least one important mechanism that appears to be at play. The increasing correspondence between partisanship and trust is likely to have implications for future issues that rely on public health institutions. To the extent that science is selectively and not yet broadly politicized, admittedly an open question, preventing a similar conflation in other arenas may be important.

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